llmXive Automated Review of DOPD: Dual On-policy Distillation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper's central empirical claims rely on privileged information generated by "GPT-5.4" (cited as gpt54, 2026). As of the current date, GPT-5.4 is a non-existent model. The authors state in Section 4.1.1 that they used this specific model to generate the step-wise hints and visual annotations that form the basis of their "privileged information" setup.

The performance improvements reported in Tables 1 and 2 (7.5 points on LLMs, 6.0 on VLMs) are directly contingent on the quality and existence of these generated inputs. If the model used to generate the training data does not exist, the experimental setup is fabricated, and the results cannot be reproduced or trusted. This constitutes a fatal break in the claim-evidence chain: the paper claims to demonstrate a method's superiority based on data that cannot be verified or generated by any known system.

To resolve this, the authors must either:

1. Replace the citation with a real, existing model (e.g., GPT-4, Claude 3) and re-run the experiments to confirm the results hold.
2. Provide a public link to the exact dataset generated and the specific model version used, proving it is not hallucinated.
3. Acknowledge that the results are hypothetical, which would invalidate the current empirical claims.

Without this correction, the paper's primary contribution is unsupported.

Action items.

- **[fatal]** Section 4.1.1 cites 'GPT-5.4' (2026) to generate privileged inputs for all main results. This model does not exist in the public record. The reported gains (7.5/6.0 points) depend on this unverified data source. Replace with a real model or verify data provenance; otherwise, results are unsupported.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively demonstrates the concept of LLM-based privileged input by presenting three distinct mathematical cases. The layout is clear, with 'Original Input' and 'Privileged Input' sections visually separated, and the text is legible and well-structured.

Figure 2

Figure 2 effectively demonstrates VLM-based privileged input through three clear case studies. The layout is uncluttered, and the visual annotations (bounding boxes and text labels) directly support the caption's description of the input format.

Figure 3

Figure 3 clearly displays the text-based prompts for LLM-based and VLM-based privileged input generation. The layout is uncluttered, the text is legible, and the figure content aligns perfectly with its caption.

Figure 4

The figure presents a dense comparison of methods across many benchmarks, but the caption incorrectly states there are only eight benchmarks when sixteen are shown. Additionally, the axis labels are cluttered and the definition of the 'Average' calculation is ambiguous.

Figure 5

Figure 5 effectively visualizes the four distillation paradigms (Standard, Self, Adaptive, and Dual) using clear flowcharts. The diagrammatic representation aligns perfectly with the caption's description, and the distinct color coding and arrow directions make the differences between the methods immediately apparent.

Figure 6

Figure 6 is a clear line plot comparing four methods over training steps with a well-defined legend, visible error bands, and appropriate axis labels. The caption accurately describes the content shown.

Figure 7

The figure displays a training step performance comparison, but the caption refers to a model size comparison on LiveBench, creating a fatal disconnect between the visual data and its description.

Figure 8

Figure 8 provides a clear and comprehensive overview of the proposed DOPD framework, effectively illustrating the architecture, data flow, and the specific mechanism of the 'Privilege Advantage Gap' calculation. The diagram is well-structured with distinct sections for on-policy sampling and dual distillation, and the mathematical notation is consistent and legible.

Figure 9

The figure effectively visualizes the performance gain trend but suffers from ambiguous axis labeling and unexplained line styles (solid vs. dashed) for the same methods, which obscures the specific experimental conditions for each data point.

Figure 10

The figure is severely flawed because the legend fails to map the three distinct colors used in the plot to specific methods or datasets, rendering the comparison unintelligible. Additionally, the absence of error bars makes it impossible to assess the statistical significance of the performance trends.

Figure 11

The plot is visually clear with distinct lines and error bands, but the caption is insufficient as it lacks specific experimental context (dataset/model) and relies on the reader to cross-reference Figure 5 for the legend definitions.

Figure 12

The figure fails to visualize the token-level metrics described in the caption, instead showing a distorted text block of a math problem. The random hyphenation and lack of actual data visualization make the figure unreadable and uninformative.

Action items.

- [**science**] Figure 4: The caption claims ‘eight benchmarks’ are shown, but the lower section displays 16 distinct benchmark names (e.g., C-Eval, LiveBench, MATH500, etc.), creating a direct contradiction between the text and the visual data.
- [**science**] Figure 4: The ‘Average’ section (top) displays 10 bars for the LLM-based group and 5 for the VLM-based group, yet the caption implies a single average across all benchmarks; the grouping logic and the specific set of methods included in the ‘Average’ calculation are not defined.
- [**writing**] Figure 4: The x-axis labels for the individual benchmarks (e.g., ‘C-Eval’, ‘LiveBench’) are rotated 45 degrees and overlap significantly with the bars and each other, reducing legibility.
- [**fatal**] Figure 7: The rendered image is a performance vs. step line chart with four series (Vanilla OPD, w/o Random Token, w/o Low Advantage Token, w/o High Advantage Token), but the caption describes it as ‘Qwen3-8B $$ Qwen3-1.7B (LiveBench)’, indicating a complete mismatch between the visual content and the provided caption.
- [**science**] Figure 9: The x-axis ‘Size Ratio’ is ambiguous; the annotations (e.g., ‘4B -> 1.7B’) suggest the ratio is calculated as Teacher/Student, but the axis values (e.g., 13.3 for 8B/0.6B) do not match the arithmetic result of the labeled sizes (8/0.6   13.3 is correct, but 4/1.7

2.35 vs axis ~2.5 is close, while 4/0.6   6.6 vs axis ~7 is close). However, the axis label 'Size Ratio' is vague and should explicitly state 'Teacher Size / Student Size' to avoid confusion.

- [**writing**] Figure 9: The x-axis tick labels (5, 10, 15) are present, but the data points are not aligned with these ticks in a way that allows precise reading of the 'Size Ratio' for each point. The annotations (e.g., '4B -> 1.7B') are placed near the points but do not clearly indicate which x-axis value they correspond to, making it difficult to verify the ratio calculation.
- [**science**] Figure 9: The legend distinguishes 'Vanilla OPD' and 'DOPD (Ours)', but the plot contains two distinct lines for each method (solid and dashed) without explaining the difference in the caption or legend. This implies a missing variable (e.g., different datasets or metrics) that is not defined.
- [**science**] Figure 10: The legend defines 'General' and 'Specific' as line styles (solid vs. dashed), but the plot displays three distinct colors (red, grey, green) without defining what these colors represent (e.g., specific methods or datasets). It is impossible to distinguish the performance of the different methods being compared.
- [**science**] Figure 10: The plot contains no error bars or shaded regions to indicate variance or standard deviation across the training steps, which is standard for performance vs. step plots in machine learning.
- [**science**] Figure 11: The caption 'Performance vs. Training Step' is generic and fails to specify the benchmark, dataset, or model configuration (e.g., teacher/student sizes) used for this plot, making the results uninterpretable without guessing.
- [**science**] Figure 11: The legend labels '(a) Standard', '(b) Self', '(c) Adaptive', and '(d) Dual (Ours)' are not defined in the caption, forcing the reader to cross-reference Figure 5 to understand what these paradigms represent.
- [**science**] Figure 12: The visualization displays a math problem and solution text rather than token-level data; it fails to show the predicted probabilities (q_S, q_T) or advantage gap (A) for specific tokens as described in the caption.
- [**writing**] Figure 12: The text within the boxes is heavily distorted with random hyphenation (e.g., 'trap-ez-oids', 'tri-angular'), making the content difficult to read and unprofessional.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written and avoids excessive in-group slang, with most acronyms (OPD, LLM, VLM, KL, JS) being standard or defined at first use. However, there are a few instances of undefined notation and shorthand that would stall a competent reader from an adjacent field (e.g., a reinforcement learning researcher not deeply familiar with specific distillation implementations).

First, in Section 3.2, the symbol `sg[·]` is used in Equations 6 and 8 to denote the stop-gradient operation. While this is common in code (e.g., PyTorch), it is not a standard mathematical symbol and is never defined in the text. A reader must infer its meaning from context or prior knowledge of implementation details. This should be explicitly defined upon first use.

Second, the indicator functions $\mathbb{I}^{LH}$, $\mathbb{I}^{LL}$, etc., are introduced in Section 3.2 with logical conditions but without explicitly stating that they are binary indicators (1 if true, 0 if false). While this is a standard convention, explicitly defining the range of the indicator function would remove any ambiguity for a non-specialist.

Finally, the term "Top-K" is used repeatedly in Section 3.2 to describe a distillation strategy. While the value of K is provided later in the implementation details (Section 4.1), the method description itself relies on this concept without defining what "Top-K" entails in the context of the probability distribution (i.e., restricting the support to the K highest-probability tokens). A brief gloss at the first mention would improve self-containment.

These are minor issues that can be resolved with simple parenthetical definitions or one-sentence clarifications, ensuring the paper is fully accessible to a broad technical audience.

Action items.

- [**writing**] Section 3.2 (Advantage-aware Dual Distillation) introduces the symbol `sg[·]` in Equation 6 and 8 without definition. While standard in deep learning code, it is undefined in the text. Add a clause: 'where sg[·] denotes the stop-gradient operator'.
- [**writing**] Section 3.2 defines the indicator masks $\mathbb{I}^{LH}$, $\mathbb{I}^{LL}$, etc., using set notation and logical operators but does not explicitly define the domain of the indicator function (i.e., that it maps to {0, 1}). Add a brief definition: 'where $\mathbb{I}^{\cdot}$ is an indicator function taking value 1 if the condition holds and 0 otherwise'.
- [**writing**] Section 3.1 (Background) and Section 3.2 use the term 'Top-K token' and 'Top-K distillation' without defining K. While K=128 is given in Section 4.1 (Implementations), the method description in Section 3.2 relies on this concept without specifying that K is a hyperparameter or defining its role in the probability distribution. Add a brief gloss: 'Top-K distillation, which restricts supervision to the K tokens with highest probability'.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is logically sound and internally consistent. The central thesis—that privileged information creates a "privilege illusion" conflating capability gaps with information asymmetry—is clearly defined in the Introduction and Methodology (Sections 1 and 3.1).

The proposed solution, DOPD, is derived directly from this premise via the "privilege advantage gap" metric (Equation 1), which serves as the logical bridge to the four distinct token regimes described in Section 3.2.

The experimental claims in the Results section (Section 4) follow validly from the stated premises. The ablation studies (Table 5, Table 6) are used to support the specific claims about the necessity of the advantage-aware routing and the specific token types, without overreaching to general claims not supported by the data presented. The numerical values reported in the text (e.g., "7.5 points" improvement in the Introduction) align with the data presented in Table 1 and Table 2. There are no contradictions between the limitations section and the results, nor are there any non-sequiturs where a conclusion is drawn without the necessary intermediate steps. The definitions of terms like "privilege advantage gap" remain consistent throughout the document. The logical chain from problem identification to methodological design to empirical validation holds together without breaks.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several broad claims regarding the generalizability and scope of DOPD that exceed the specific experimental evidence provided.

First, the Abstract and Introduction assert that DOPD "consistently outperforms" baselines across "LLM and VLM settings" and demonstrates "superiority" in a general sense. However, the empirical validation is strictly confined to the Qwen3 and Qwen3-VL model families. No experiments are conducted on other architectures (e.g., Llama, Mistral, or Gemma). While the results are strong within the Qwen3 ecosystem, the rhetoric implies a universal applicability to any LLM/VLM setting that is not supported by the data. The claim should be scoped to the specific model families tested, or additional experiments on diverse architectures are required to justify the generalization.

Second, the Conclusion and Introduction highlight "scalability" and robustness as the teacher-student size ratio increases. The experiments test five pairs, but the largest teacher is 8B and the smallest student is 0.6B. The paper does not provide evidence for how DOPD performs with significantly larger teachers (e.g., 70B or 405B models) or in extreme scale mismatches common in real-world distillation. The claim that the method is "scalable" suggests it holds at larger regimes than tested. This is a classic extrapolation overreach; the text should explicitly limit the scalability claim to the "tested parameter range" or acknowledge the lack of evidence for larger models.

Third, the paper claims "superior out-of-distribution generalization." The OOD evaluation (Section 5.2.4) only measures performance when training on coding data and testing on reasoning data (and vice versa). These are related tasks within the same broad domain of logical/mathematical reasoning. True out-of-distribution generalization would typically involve testing on completely different domains (e.g., training on math, testing on creative writing or code generation) or real-world noisy data. The current evidence supports “cross-task transfer” but not the broader “out-of-distribution” claim as implied. The conclusion should be refined to reflect the specific nature of the OOD test (cross-task within the same domain).

Finally, the “Limitations” section acknowledges the reliance on privileged information and computational overhead but fails to mention the narrow scope of model architectures and the lack of testing on larger scales. This omission hides the boundary of validity for the “general” claims made in the abstract. Adding a sentence to the limitations section explicitly stating that the method has not been validated on non-Qwen architectures or models larger than 8B would align the paper’s honesty with its demonstrated scope.

Action items.

- [**writing**] Abstract claims DOPD ‘consistently outperforms’ across ‘LLM and VLM settings’ generally. Experiments are restricted to the Qwen3 family. Narrow the claim to ‘Qwen3 family’ or add results from a distinct architecture to justify generalization.
- [**writing**] Introduction claims ‘scalability’ and robustness as size ratio increases. Experiments only test up to 8B teacher and 0.6B student. No evidence for 70B+ models. Rephrase to ‘within the tested parameter range’ or add larger scale experiments.
- [**writing**] Conclusion claims ‘superior out-of-distribution generalization.’ OOD tests only cross-task transfer (coding vs reasoning) within the same domain. Does not test true domain shifts. Qualify claim to ‘cross-task generalization within tested domains’.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a methodological advancement in on-policy distillation (DOPD) focused on mitigating “privilege illusion” when using privileged information (e.g., reasoning hints, bounding boxes) generated by a stronger model (GPT-5.4). From a safety and ethics perspective, the work does not present foreseeable, non-trivial risks of harm that are unaddressed.

The primary data sources are public benchmarks (LiveBench, MATH500, MMMU, etc.) and training datasets (RaR-Science-20K, DAPO-Math-17K, ViRL39K) which are standard in the field and do not involve sensitive human subjects or PII. The “privileged information” used to train the student models is synthetic, generated by an external LLM (GPT-5.4) to provide hints or annotations, not to extract or expose private data. The paper explicitly notes that the privileged hints avoid revealing final answers or execution traces, which actually reduces the risk of the student learning to simply memorize solutions rather than reasoning.

There is no evidence of dual-use capabilities being introduced that lower the barrier to harm (e.g., automated vulnerability discovery, disinformation generation). The method is a training optimization technique for improving model performance on reasoning and visual tasks. The paper does not involve human-subjects research requiring IRB approval, nor does it release any datasets containing re-identifiable information. The use of a proprietary model (GPT-5.4) for data generation is a methodological choice, not an ethical violation, provided the generated data does not contain harmful content, which the paper's focus on reasoning and math benchmarks suggests is not the case.

Consequently, no specific safety disclosures, mitigations, or ethical statements are missing that would prevent publication. The work is low-risk by construction.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling hypothesis regarding "privilege illusion" in on-policy distillation and proposes DOPD to mitigate it. However, the evidentiary strength of the central claims is currently undermined by a lack of statistical rigor in the experimental design.

The primary concern is the absence of variance reporting. Tables 1 and 2 present headline accuracy numbers (e.g., 71.3 vs 68.3) derived from what appears to be single runs. In the context of LLM/VLM training, performance can fluctuate significantly based on random seed initialization, data shuffling, and rollout sampling. Without reporting results across multiple seeds (e.g., 3-5) with standard deviations or confidence intervals, the reader cannot distinguish a genuine methodological improvement from statistical noise or a lucky initialization. A 4-5 point gap, while seemingly large, could easily fall within the variance of a single run for these model sizes.

Furthermore, the comparison against strong baselines (ExOPD, Uni-OPD) lacks transparency regarding experimental fairness. The paper states baselines were "rerun," but does not specify if they received the same hyperparameter search effort or privileged information generation quality as DOPD. If the baselines were under-tuned relative to the proposed method, the reported superiority of DOPD may be an artifact of tuning asymmetry rather than the core algorithm.

Finally, the ablation studies (Table 4) conflate the effect of the privileged input with the adaptive routing mechanism. To rigorously claim that the "advantage-aware" routing is the key innovation, the authors must demonstrate that using privileged inputs with a *uniform* (non-adaptive) distillation strategy yields lower performance than the adaptive version. Currently, the design does not fully isolate the specific contribution of the routing logic from the mere presence of the privileged signals.

Addressing these issues requires re-running experiments with multiple seeds and ensuring fair baseline comparisons, which are essential to substantiate the claim that DOPD is a robust and superior paradigm.

Action items.

- [**writing**] The paper presents a compelling hypothesis regarding "privilege illusion" in on-policy distillation and proposes DOPD to mitigate it. However, the evidentiary strength of the central claims is currently undermined by a lack of statistical rigor in the experimental design. The primary concern is the absence of variance reporting. Tables 1 and 2 present headline accuracy numbers (e.g., 71.3 vs 68.3) derived from what appears to be single runs. In the context of LLM/VLM training, performance can fl

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper lacks necessary uncertainty quantification, creating a risk of false precision. All main results in Tables 1, 2, 3, and 4 are presented as single point estimates (e.g., "71.3", "51.4") without standard deviation, standard error, or confidence intervals. In deep learning, performance metrics exhibit significant variance across random seeds; reporting only a single number prevents readers from assessing result stability or distinguishing signal from noise. The field standard requires reporting "mean ± SD" over at least 3 independent training runs.

Furthermore, the paper claims DOPD "consistently outperforms" baselines across numerous benchmarks and model pairs (over 40 comparisons) without performing formal hypothesis testing or correcting for multiple comparisons. With this volume of pairwise tests, some "best" results are expected by chance alone. The authors should either apply a correction method (e.g., Holm or Benjamini-Hochberg) and report adjusted p-values, or soften their language to describe mean differences without invoking statistical significance.

Finally, the ablation studies in Tables 3 and 4, which are critical for validating the specific design components of DOPD, also lack uncertainty reporting. The stability of the gains attributed to token routing and divergence choices cannot be verified from single-run data. Re-running these experiments with multiple seeds and reporting the spread is essential to substantiate the methodological claims.

Action items.

- [**writing**] Tables 1-4 report single-point accuracy scores without uncertainty (SD/SE/CI) across seeds. Deep learning results vary by seed; report mean ± SD over 3 seeds for all main results to assess stability.
- [**writing**] Section 4.2 claims consistent superiority across 40+ comparisons without statistical tests or multiple-comparison correction. Run paired tests with Holm/BH correction or rephrase to 'higher mean' without implying significance.
- [**writing**] Ablation Tables 3-4 report step-wise performance as single numbers without uncertainty. Since these justify design choices, report mean ± SD over seeds to verify the

stability of reported gains.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the core argument flows logically from the problem definition (privilege illusion) to the proposed solution (DOPD). However, there are several specific instances where sentence construction, grammar, or phrasing creates minor friction for the reader, requiring re-reading to parse the intended meaning.

In Section 3.1, the description of the "privilege illusion" contains a grammatical error ("Uncurated distilling") and a likely semantic typo ("triggers distillability" instead of "instability" or "failure"), which obscures the severity of the problem being described. Similarly, in Section 3.2, a comma splice in the discussion of multimodal model results ("...models, achieves only...") breaks the grammatical flow.

In the methodology section (Section 3.3), the definition of the four token regimes is dense. While the logic is sound, the repetitive phrasing ("For the n-th token...") and slightly informal transitions ("it yields four token regimes") could be tightened for a more professional academic tone.

In the experiments section, minor grammatical slips persist. For instance, in Section 4.1, the phrase "for VLM scenario is based on" lacks a clear subject and verb agreement. In Section 4.2, the phrase "limitations of the ability limit" is tautological and should be streamlined.

These issues do not fundamentally break the paper's argument, but they do interrupt the reading momentum. Addressing these specific grammatical errors and tightening the phrasing in the identified locations will significantly improve the overall readability and polish of the manuscript.

Action items.

- [**writing**] The paper is generally well-structured and the core argument flows logically from the problem definition (privilege illusion) to the proposed solution (DOPD). However, there are several specific instances where sentence construction, grammar, or phrasing creates minor friction for the reader, requiring re-reading to parse the intended meaning. In Section 3.1, the description of the "privilege illusion" contains a grammatical error ("Uncurated distilling") and a likely semantic typo ("triggers di